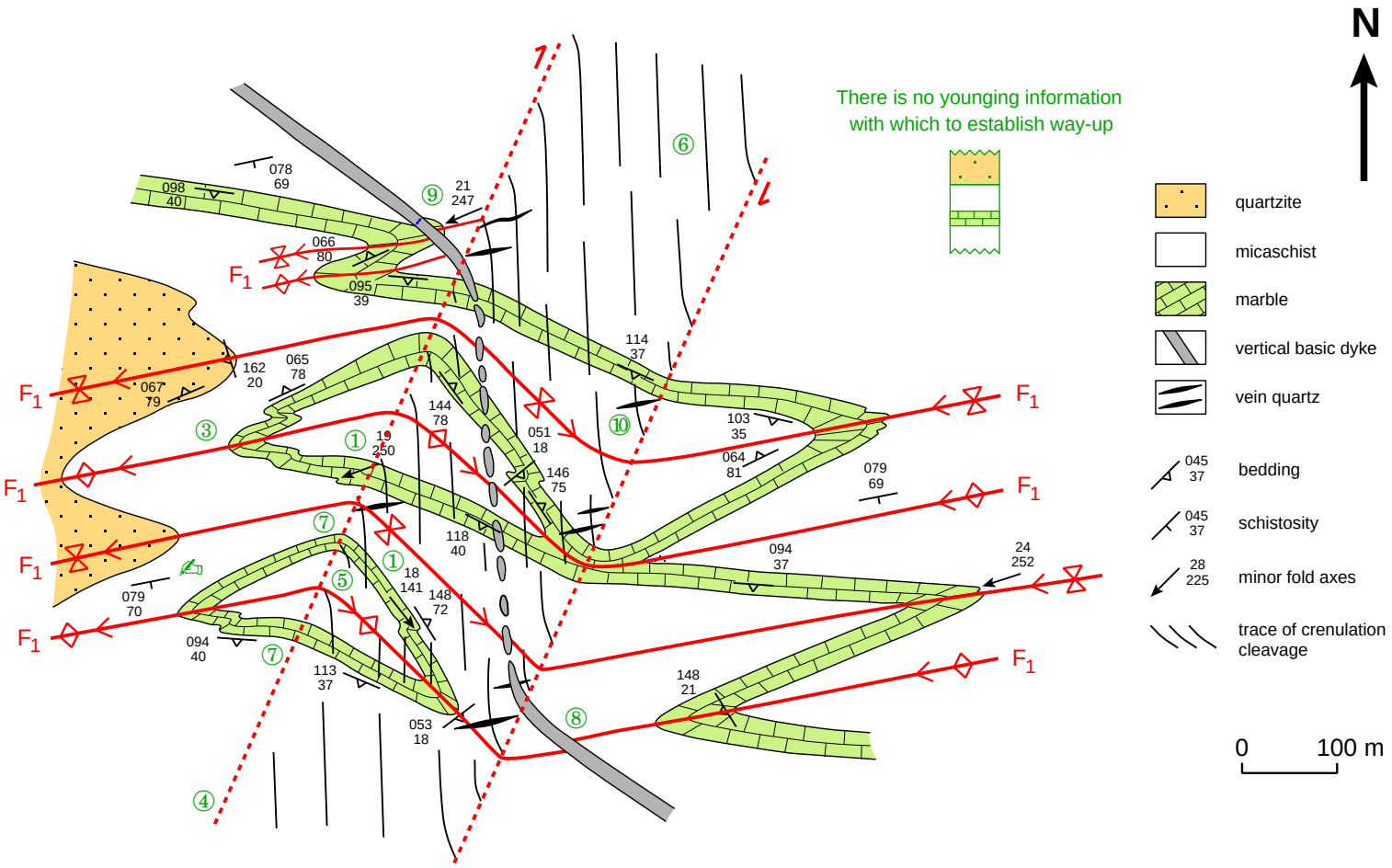
SHEAR ZONE EXERCISE



1 The minor folds are not in constant orientation and so are F_1.

2 The schistosity is axial planar to the minor folds and so is S_1.

3 The axial traces are located by the folded lithological contacts and changing the minor fold asymmetries. They are folded and hence are F_1 folds.

4 The F_2 “axial traces” are not folded implying that they have not been affected by any later folding event.

5 D_2 has reoriented the F_1 folds to produce a fold interference pattern.

6 The crenulation cleavage (S_2) is only developed between the two F_2 axial traces and it is not axial planar to the F_2 folds. It is, however, consistent with the orientation of a cleavage generated within a shear zone which has the two F_2 “axial traces” as its boundaries. The orientation and curvature of S_2 implies that the slip vector has a dextral strike-slip component while, in principle, the fact that the angle between S_2 and the shear zone boundaries is less than 45° suggests transpression, that is, that the slip vector also has a reverse dip-slip component.

7 The different limbs of the F_1 folds were reorientated by different amounts during the D_2 shearing, which is also consistent with there being a dip-slip component to the D_2 slip vector.

8 The dyke crosscuts the F_1 axial traces but is boudinaged across the D_2 shear zone implying that it was intruded between D_1 and D_2. The boudinage and apparent offset of the dyke across the shear zone is consistent with a dextral strike-slip component to the slip vector, while the fact that the apparent offset of the dyke is larger than that of the fold traces is consistent with there being a dip-slip component.

9 The dyke opened normal to its margins.

10 The orientation of the vein quartz suggests that these veins are tension gashes formed during D_2 shearing.